

2023



The Future of Property Inspections

Bees360's Drone-Assisted Property Inspection Solution on AWS Helps Insurers Save 30% on Loss Adjustment Expenses

Learn how technology startup Bees360 built a drone-assisted home inspection solution using AWS.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcomes](#) | [AWS Services Used](#)



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30%

reduction in loss adjustment expenses

3%

reduction in loss ratios

2 months

to build the complete solution

Achieved

SOC 2 Type 2 compliance



Breakthrough

data scalability in the hundreds of terabytes

Overview

In the United States, [88 percent of homeowners](#) have homeowner's insurance to cover property losses caused by natural disasters, such as hurricanes, floods, and hail. For over a century, property insurance and claims inspections have been conducted manually. Due to the specialized nature of property inspections, the process is costly and requires trained and certified professionals. Additionally, injuries sometimes occur due to accidents during roof inspections. Insurance companies are seeking new ways to conduct property inspections to enhance safety, improve the customer experience, and improve the efficiency of underwriting and claims processes.

[Bees360](#) is a technology startup that aims to revolutionize property inspections in the insurance industry. Leveraging its drone and artificial intelligence (AI) technologies, Bees360 replaces traditional, manual operations with modern technology that delivers precision in every pixel and positions insurance companies to be more innovative.

Bees360 built their platform on Amazon Web Services (AWS). By using a range of AWS serverless and managed services, Bees360 provides reliable storage and processing of various types of structured data in property inspections for homeowner's insurance. The comprehensive data security and compliance offered by AWS helps Bees360 provide products and services with critical information protection built into their process.



As a technology startup, Bees360 has extensive technical expertise in fields such as drones and AI. Its solutions are specifically designed to provide insurance companies with fast, reliable, and accurate drone-assisted property inspections during underwriting and claims processes.

“The introduction of unmanned aircraft management regulations by the Federal Aviation Administration (FAA) has presented us with new business opportunities. As the number of drones increases, our team has successfully proven that drones can be used as a new tool for insurance inspections,” says Andy Liu, CEO at Bees360. “For this purpose, we recruited certified drone pilots as our frontline property inspectors, supported by a comprehensive certification program. More importantly, we needed to build a cloud-based solution to handle tasks such as assigning inspections to pilots, collecting and processing high resolution image data, and presenting results in a highly customizable format.”

To meet the stringent data security requirements of the insurance industry and to effectively handle massive data processing in insurance underwriting and claims investigations, Bees360 needed to carefully consider the security and stability of data when selecting a cloud service provider. After comparing multiple cloud service providers, Bees360 ultimately chose to adopt AWS to build its service backend and establish a business solution on the cloud. This cloud-based solution helps with the processing of inspection data and facilitates the exchange of claim and underwriting information with insurance companies.

“Our clients are large, top-ranked insurance companies in North America, and their insurance claims and underwriting processes require reliable data security and privacy protection solutions. Each client requires detailed security response documents from us, describing in detail how we meet the corresponding security requirements,” says Liu. “In this process, the team at AWS has been a critical partner due to their rich experience in serving financial insurance clients, mature security products, and compliance certifications. Building our business on AWS was a key factor in winning client confidence.”

**Solution | Reducing Loss Adjustment Expenses by 30%
Decreasing Underwriting Inspection Turnaround Time by 40%**



Using Amazon S3 and AWS Lambda

In May 2023, with the assistance of the AWS team, Bees360 deployed its infrastructure and the insurance claim and underwriting information exchange solution to the cloud. During this process, the team implemented multiple serverless managed services to reduce the complexity of future operations and improve the solution's flexibility. For the application's container orchestration, Bees360 uses [Amazon Elastic Kubernetes Service](#) (Amazon EKS), a managed service to run Kubernetes in the AWS Cloud and on-premises data centers, facilitating efficient resource allocation during peak periods of claim inspections after disasters, such as hurricanes. To meet the high concurrency demands of drone pilots, Bees360 uses [Elastic Load Balancing](#) (ELB), which automatically distributes incoming application traffic across multiple targets and virtual appliances. The Bees360 development team uses [AWS Lambda](#)—a serverless, event-driven compute service—to preprocess house inspection images stored in [Amazon Simple Storage Service](#) (Amazon S3), an object storage service offering cutting-edge scalability, data availability, security, and performance. This preprocessing verifies that the images meet the requirements of the detection models in terms of specification and resolution.

“Embedding various serverless managed services from AWS in our business has simplified the construction process, helping us to achieve our business goals with minimal code,” said Sunnie Wu, Chief Operating Officer. “It is very convenient.”

Bees360's solution also includes AWS services for security and compliance, such as [AWS Key Management Service \(AWS KMS\)](#), which is used to create and control keys used to encrypt or digitally sign data. Using AWS KMS, sensitive data between applications can be encrypted for secure transmission. For advanced security, Bees360 uses [Amazon GuardDuty](#), a threat detection service that continuously monitors AWS accounts and workloads, as well as [Amazon Inspector](#), an automated vulnerability management service that continually scans AWS workloads for software vulnerabilities and unintended network exposure.

“In both the underwriting and claims processes, the risk inspection data exchanged between Bees360 and the insurance company is fully encrypted, both during static storage and transmission,” says Liu.

In the context of insurance underwriting and claims property inspections, reliable data management and analysis are crucial. To store various types of structured data, Bees360 uses [Amazon Relational Database Service \(Amazon RDS\)](#)—a collection of managed services that makes it simple to set up, operate, and scale databases in the cloud. For data analysis, Bees360 chose [Amazon OpenSearch Service](#)—an open-source, distributed search and analytics suite—to provide fast and scalable full-text search capabilities for insurance underwriting and claims data, thereby accelerating the operational efficiency of insurance companies.

Bees360 built the complete business solution on AWS in two months, enabling certified drone pilots to control and transmit data through the BeesPilot App. The solution processes the property inspection images in the cloud and delivers the inspection reports to insurance companies through a secure data channel. Currently, this solution has been leveraged by insurance companies in 48 states and has been successfully applied in pre-binding underwriting process and post-disaster claims. Combining the drone-assisted inspection solution with the security features of AWS, Bees360 has successfully achieved SOC 2 Type 2 high-standard data security compliance. This significant milestone has gained the trust of their customers, including the top 10 insurance companies in the United States, and has encouraged other insurance companies to explore drone-assisted inspection solutions as a replacement for traditional boots-on-roof inspections.

By rapidly processing the imagery property data and securely exchanging it with insurance companies, Bees360 has enabled insurance companies to significantly reduce cost and increase efficiency of claims and inspections. “Our in-house developed BeesPilot App can directly store the property imagery cs to Amazon S3 and perform preliminary processing in a serverless manner through AWS Lambda. Our

self-developed AI model for comparison and review, the application ultimately presents a unified report style to insurance companies, facilitating quick claims settlement for homeowners,” says Liu.

In addition, the elastic and reliable infrastructure provided by AWS helps Bees360’s technical team to better handle the company’s rapid growth. With the elastic capabilities of Amazon S3 and Amazon RDS, Bees360 can easily scale its storage services from a few terabytes to hundreds of terabytes, accommodating the storage of numerous high-resolution images per property insurance policy as well as structured data storage for automated detection reports for underwriting and claims.

Outcome | Rolling Out Updates Using Amazon EKS and Scaling Up Using Amazon SageMaker

In terms of business continuity, Bees360 uses Amazon EKS to host Kubernetes services, facilitating rolling updates for the Bees360 platform. This service enables Bees360 to deploy software releases more frequently and deliver new features to drone pilots to improve their user experience and operational efficiency.

Looking ahead, Bees360 will continue to use AWS as the backbone for their drone-assisted property inspection and AI innovation. By introducing [Amazon SageMaker](#)—which is used to build, train, and deploy machine learning models for any use case—Bees360 aims to gradually migrate the AI model training used in property inspections from local data centers to the more secure cloud environment. This migration will help the company to prepare for scaling up property inspections and expanding into diversified business areas, such as real estate appraisal market, in the future.

About Bees360

Bees360 is a pioneering AI and data firm specializing in swift and dependable drone-assisted property inspections for insurance carriers during underwriting and claims processes. With a robust network, the company has the largest collection of drone-assisted property inspectors in the United States, with 3,500 Bees360-certified, FAA-licensed drone pilots spread across all 50 states. The company’s dynamic team consists of insurance industry veterans, elite data scientists, and top-tier product developers.

AWS Services Used

Amazon Elastic Kubernetes Service

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run Kubernetes in the AWS cloud and on-premises data centers.

[Learn more »](#)

Elastic Load Balancing

Elastic Load Balancing (ELB) automatically distributes incoming application traffic across and virtual appliances in one or more Availability Zones (AZs).

[Learn more »](#)



AWS Lambda

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend service without provisioning or managing servers.

[Learn more »](#)

Amazon S3

Object storage built to retrieve any amount of data from anywhere.

[Learn more »](#)

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